Analysis of Variance

One-Way ANOVA

Exercise: The times required by three workers to perform an assembly-line task were recorded on five randomly selected occasions. Here are the times, to the nearest minute.

Hank	Joseph	Susan
8	8	10
10	9	9
9	9	10
11	8	11
10	10	9

Solve with ANOVA with significance level: 0.05

One-Way ANOVA

Solution:

One-Way ANOVA

Exercise2: Data on Scholastic Aptitude Test (SAT) scores are published by the College Entrance Examination Board in *National College-Bound Senior*. SAT scores for randomly selected students from each of four high-school rank categories are displayed in the following table.

Top Tenth	Second Tenth	Second Fifth	Third Fifth
528	514	649	372
586	457	506	440
680	521	556	495
718	370	413	321
	532	470	424
			330

Solve with ANOVA with significance level: 0.1

One-Way ANOVA

Solution:

One-Way ANOVA

Exercise3:

Four brands of flashlight batteries are to be compared by testing each brand in five flashlights. Twenty flashlights are randomly selected and divided randomly into four groups of five flashlights each. Then each group of flashlights uses a different brand of battery. The lifetimes of the batteries, to the nearest hour, are as follows.

Brand *A*	Brand *B*	Brand *C*	Brand *D*
42	28	24	20
30	36	36	32
39	31	28	38
28	32	28	28
29	27	33	25

Solve with ANOVA with 5% significance level

One-Way ANOVA

Solution: